

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level (K1-Remembering, K2- Understanding, K3 – Applying, K4 – Analyzing, K5 – Evaluating, K6 - Creating) of question or expected answer as per OBE or the course outcome (CO) on which the question is based is mentioned in front of each question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(CO/Level)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Identify the correct option from the following options that show the major consideration in good electrical machine design?		
a) Cost	CO1/K1	1
b) Durability		
c) Compliance of performance criteria		
d) All of the above		
2 Pick out suitable option from the following options that primarily ensure that the temperature rise is kept within acceptable limits.		
a) Good efficiency of the machine	CO1/K1	1
b) Proper cooling and ventilation		
c) Proper saturation level		
d) Good insulation		
3 Recognize the correct formula for heat radiated from a body as per Stefan's law.		
a) $H = 5.67ne[(T_1^4/100-T_2^4/100)] W/m^2$		
b) $H = 4.67ne[(T_1^4/100-T_2^4/100)] W/m^2$	CO2/K1	1
c) $H = 5.67ne[(T_2^4/100-T_1^4/100)] W/m^2$		
d) $H = 5.67ne[(T_1/100)^4-(T_2/100)^4] W/m^2$		
4 In a lap winding of a "p" pole and "m" winding dc machine, Identify the correct formula for the number of brushes.		
a) mp	CO2/K1	1
b) m/p		
c) 2m/p		
d) m/2p		

- 5 **When** the voltage and frequency of a three phase induction motor is halved,
- air gap flux is halved.
 - slip at rated torque increases greatly.
 - the maximum torque is halved.
 - the maximum torque remains the same.
- CO3/K1 1
- 6 **Identify** the correct statement from the following about the effect of reducing the frequency of AC supply with a corresponding reduction in applied voltage on the three-phase induction motor.
- It has no effect on the magnitude of starting current.
 - Reduce magnitude of starting current.
 - Increase the magnitude of starting current.
 - All of the above.
- CO3/K1 1
- 7 **Pick out** the correct option from the following when the stator of a three-phase, 4-pole squirrel cage induction motor rated for 1450 rpm at 50 Hz is rewound to have six poles without any change made on the rotor. This motor would then
- run at the same speed with higher torque rating
 - operate with speed slightly below 1000 rpm at 50 Hz
 - fail to develop any torque
 - operate with speed slightly above 1800 rpm at 50 Hz
- CO4/K1 1
- 8 **When** the air gap in an induction motor is increased, then
- the magnetizing current of the rotor will decrease.
 - the power factor will decrease.
 - the speed of motor will increase.
 - the windage losses will increase.
- CO4/K1 1
- 9 **Identify** the correct option from the following that is NOT a desirable property in transformer oil.
- Low evaporation
 - Low thermal conductivity
 - Low cost
 - High insulation resistance
- CO5/K1 1
- 10 **Pick out** the correct option from the following that is NOT a valid method of cooling of transformer.
- Oil Natural Water Forced
 - Oil Natural Air Forced
 - Oil Forced Water Forced
 - Water Natural Oil Forced
- CO5/K1 1

11 Recognize the correct material from the following: the magnetic frame of a large transformer is built up of,

- a) nickel alloy steel
- b) chrome sheet steel
- c) hot called silicon steel
- d) cold rolled grain-oriented silicon steel

CO6/K1 1

12 Pick out the correct option from the following: the bushings in transformers used for

- a) They are used to cool the transformer oil.
- b) They are used to filter contaminants from the transformer oil
- c) They are used to connect the transformer to the power grid.
- d) They are used to insulate the transformer leads as they come out through the tank

CO6/K1 1

Q.2 Solve the following.

12

- A) **Explain** the limitations considered with designing the electrical machine.
- B) **Discuss** the insulating materials class based on thermal consideration.

CO1/K2 6

CO1/K2 6

Q.3 Solve the following.

12

- A) **Show** the simplex lap-wound armature with a winding table of a 4-pole generator with 24 conductors.
- B) **Illustrate** a design procedure for choke (coil) design with relevant formulae.

CO2/K3 6

CO2/K3 6

Q.4 Solve Any Two of the following.

12

- A) **Illustrate** the design procedure of stator of squirrel cage induction machine with the help of formulae.
- B) **Compute** the main dimensions of the of a 5 h.p., 400 V, 3-phase, 4-pole squirrel cage induction motor using star-delta starter. Assume the following data:

CO3/K3 6

Average flux density in the air gap = 0.46 Wb/m²
 Ampere conductors per meter of armature periphery = 22000
 Full load efficiency = 83%
 Full load p.f. = 0.84 lag
 Winding factor = 0.955
 Stacking factor = 0.9
 Current density = 4 A /mm²
 No. of slots per poles per phase = 3

CO3/K3 6

$$L/t = 1.5$$

- C) **Illustrate** the procedure of computation of losses in rotor bars and end ring of inductor machine.

CO4/K3

6

Q. 5 Solve Any Two of the following.

12

- A) The thermal conductivity of assembled armature laminations is 20 times as great along the direction of laminations as in the direction across the laminations.

Calculate the loss that will be conducted across the laminations in a stack 40 mm thick and 6000 mm² in cross-section with a difference of 20°C. Given that a difference of 5°C will cause 25 W to be conducted through a cross-section of 2500 mm² in area and 20 mm thick measured along the laminations.

CO5/K3

6

- B) A 50 MVA turbo-alternator has a total loss of 1500 kW. **Calculate** the volume of air required per second and also the fan power if the temperature rise in the machine is to be limited to 30 °C. The other data given is;

Inlet temperature of air = 25 °C,

CO5/K3

6

Barometric = 760 mm of mercury,

Pressure = 2 kN/m²,

Fan efficiency = 0.4.

- C) **Show** various types of ventilating systems in accordance with the provision of cooling ducts and how the air passes over the heated part of the machine.

CO5/K3

6

Q. 6 Solve Any Two of the following.

12

- A) **Illustrate** the types of tests performed on transformers as per the Standards for estimation of leakage reactance, resistance of winding, no load current, and losses.

CO6/K3

6

- B) **Determine** dimensions of magnetic frame of a 100 kVA, 11 kV/415 V, 50 Hz, three-phase delta/star core type distribution transformer. Assume flux density = 1.2 T, current density = 2.5 A/mm², window space factor = 0.28, and constant K = 0.45 for three-phase core-type distribution transformers. Use a four-step core.

CO6/K3

6

- C) **Illustrate** the step-by-step design procedure of winding for a three-phase power transformer with relevant formulas.

CO6/K3

6

*** End ***